QK-Norm

Normalize queries and keys so attention logits stay bounded

Zhipu AI & Tsinghua · GLM-4.5 (after Henry et al. 2020, Dehghani et al. 2023)

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Muon-era MoEs all carry a logit control

Model	Attention	Optimizer	Logit control
DeepSeek-V3	MLA	AdamW	none
Kimi K2	MLA	Muon	QK-Clip (deck 65)
GLM-4.5	**GQA**	**Muon**	**QK-Norm**
GLM-5	MLA	Muon	Muon Split
DeepSeek-V4	CSA / HCA	Muon	RMSNorm on q and KV

GLM-4.5 trains a 355B MoE with Muon and bounds logits with QK-Norm

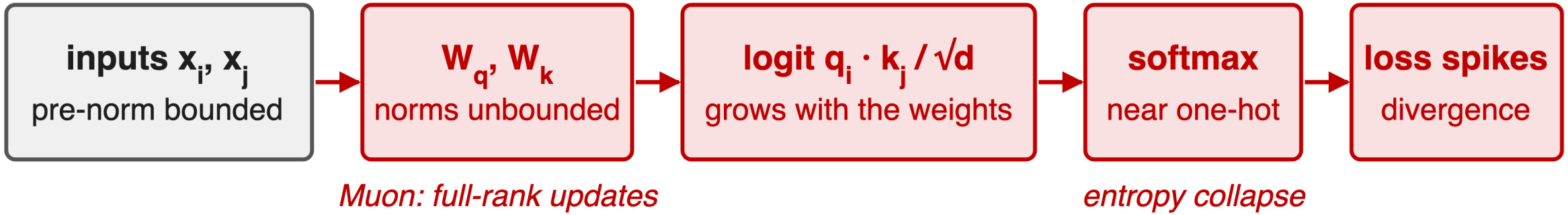
Muon lets attention logits run past 1,000

> 1,000

max attention logit of a 53B-total MoE trained with vanilla Muon: loss spikes, occasional divergence

Kimi K2 tech report, arXiv 2507.20534, §2.1 and Fig. 2 (mid-scale run, 9B activated / 53B total)

Nothing in the attention block bounds Wq, Wk



$$|q_i \cdot k_j| \le \|x_i\| \, \|x_j\| \, \|W_q\| \, \|W_k\|$$

8B ViT with Adam: logits above 50,000, one-hot attention, divergence

Lower LR, soft-caps and clipping act too late

- **Lower the LR** (ViT-H: 1e-3 → 4e-4): stable, **but loses accuracy**
- **Soft-cap logits** (Gemma 2): **q·k keeps growing** under the cap
- **Clip Wq, Wk** (QK-Clip, deck 65): **needs per-head max logits**

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All three react after q·k has grown; none removes the weight norm

Insight: RMSNorm cancels the weight norm

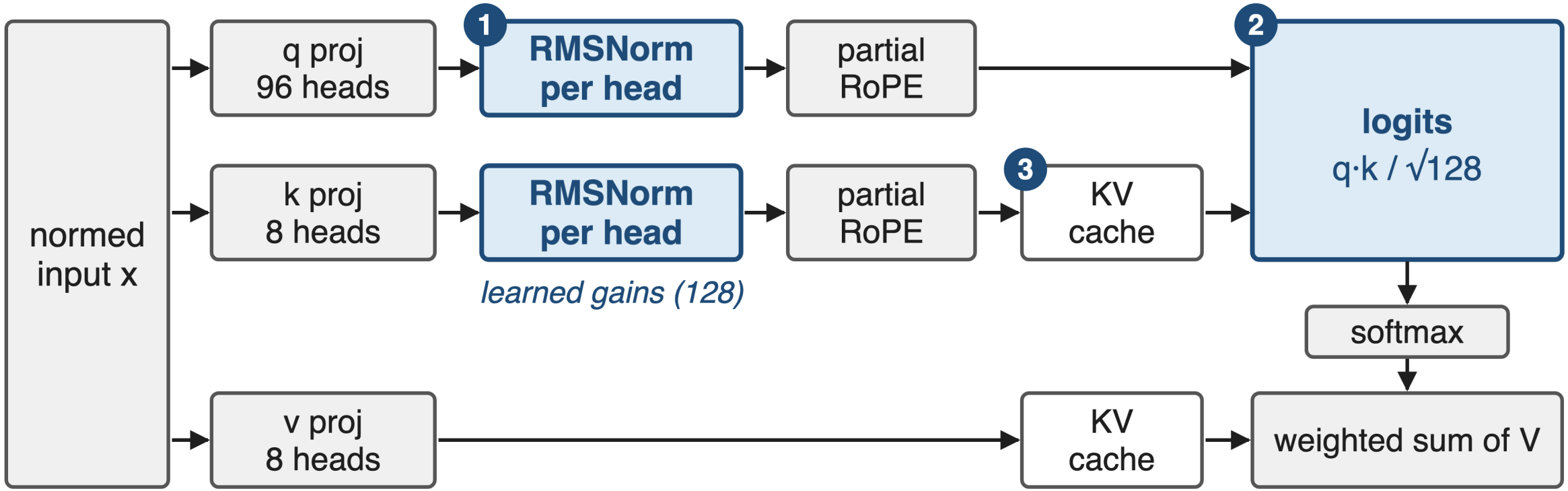
- `RMSNorm(c·q) = RMSNorm(q)` for any scale $c > 0$
- Apply it to q and k **after** the projection

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The weight norms drop out of the logit: bounded by construction

QK-Norm sits between projection and RoPE

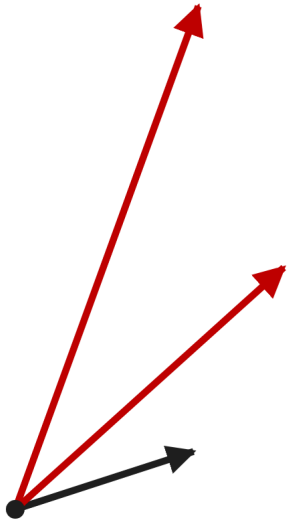


GLM-4.5 attention as implemented in transformers' modeling_glm4_moe.py; values are not normalized

❶ Per-head RMSNorm puts q and k on a sphere

raw q, k

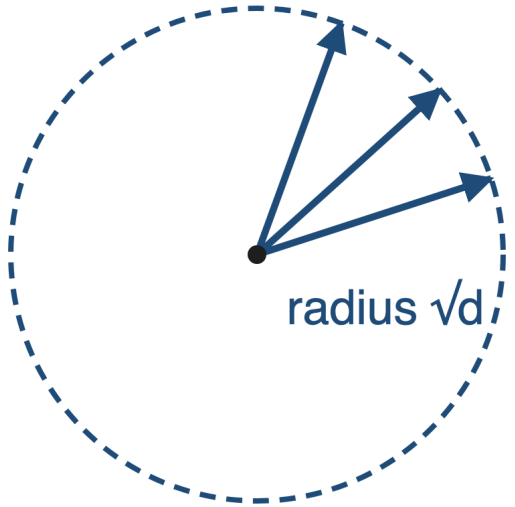
length grows with $\|W_q\|$



RMSNorm

per head

after QK-Norm



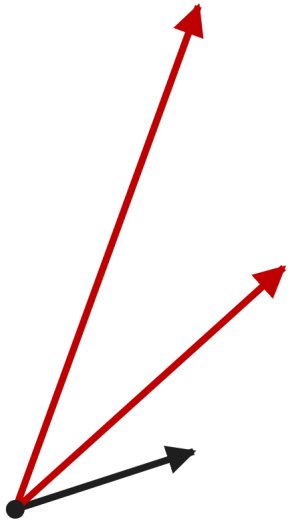
(unit gains)

Every 128-dim q and k head ends at length √128 × gain, whatever the weights

❷ Logits bounded by √d times the gains

raw q, k

length grows with $\|W_q\|$

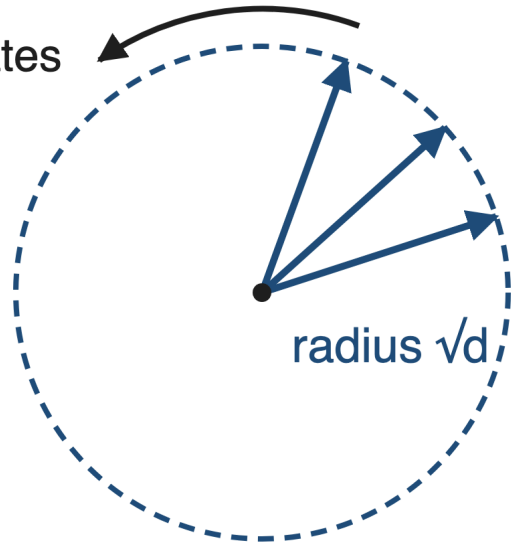


RMSNorm

per head

after QK-Norm

RoPE rotates



(unit gains)

max logit ≤ $\sqrt{d} \cdot g_q \cdot g_k$

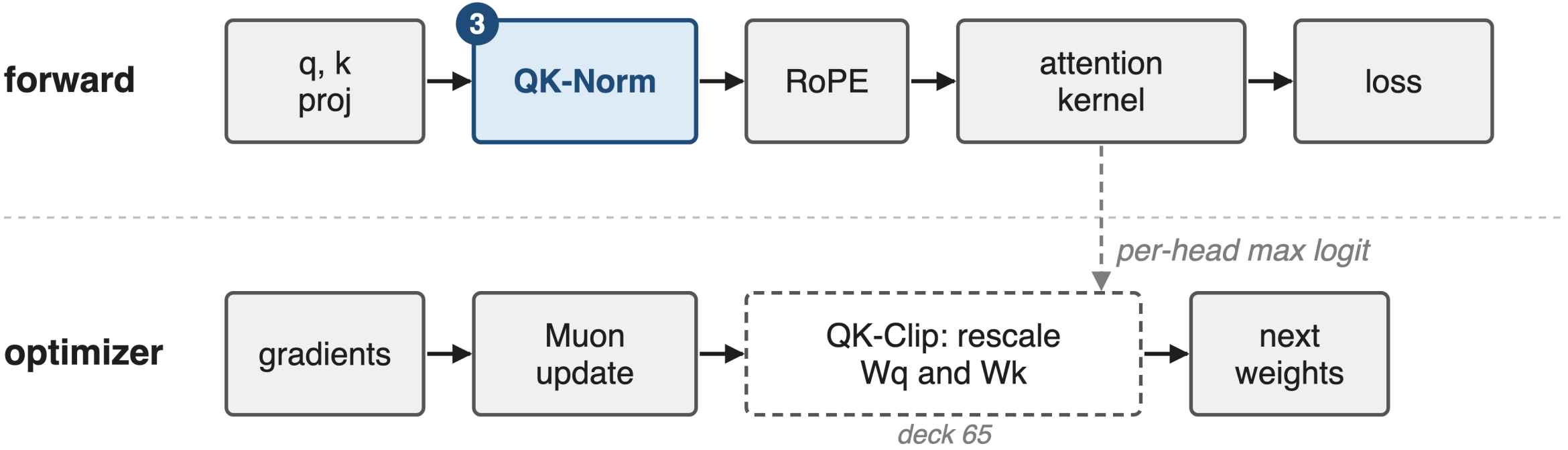
d = 128, unit gains

max logit ≤ 11.3

g: max learned gain

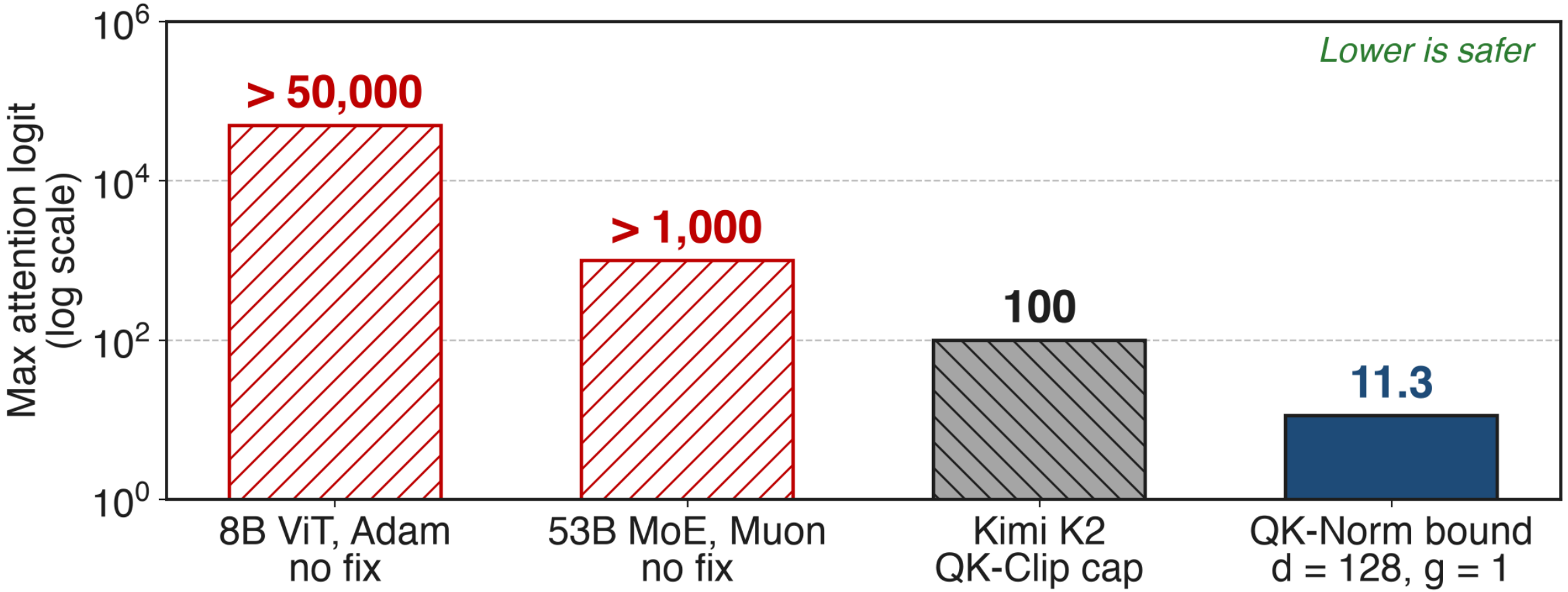
Sharpness becomes a learned temperature: 2 × 128 gains per layer

❸ A forward-pass op, not an optimizer hook



No statistics, no weight surgery; cached keys are already normalized

Bound: 11.3 × gains, versus > 1,000 unchecked



Measured bars are lower bounds from different models (ViT-22B App. B; K2 Fig. 2). QK-Norm: analytic bound at unit gains

Runs that diverged without it now train

- **8B ViT**: diverged at LR 1e-3; with QK norm 1e-3 stays stable
- **1.2B LM** trains at LR **0.3** with qk-layernorm
- **Chameleon-7B** without it diverges ~**20%** into an epoch

DeepSeek-V4 drops QK-Clip: RMSNorm on q and KV entries suffices

QK-Norm cannot reach MLA's absorbed keys

GQA
GLM-4.5



per-head key $k_h = W_{UK,h}\, c$ is never formed
no k to normalize

MLA decode
DeepSeek-V3, K2



MLA models bound logits elsewhere: QK-Clip (K2), Muon Split (GLM-5)

Cheap, but not a complete stability fix

- **256** gains per layer
- One norm per head, before RoPE
- No kernel or optimizer change

- **Soft bound**: gains can still grow
- **Output logits**: still need z-loss
- **Not alone**: OLMo 2 also reorders norms

GLM-4.5 publishes no ablation or logit curves for QK-Norm

Summary

- Muon grows the Wq, Wk norms: max logit **> 1,000** unchecked
- **Insight**: RMSNorm after the projection cancels the weight norm
- Per-head norm before RoPE: logit ≤ $\sqrt{d} \times$ gains = **11.3** at init
- Needs materialized keys: GQA yes, MLA no (deck 65)

https://arxiv.org/abs/2508.06471